



R.B.INSTITUTE OF MANAGEMENT STUDIES(RBIMS) MCA SEMESTER-3

SUBJECT NAME: : Design and Analysis of Algorithms
SUBJECT CODE: : MC03094011

Assignment-2

Issue Date:17/09/2026 Submission Date:22/09/2026 1. Explain Rod Cutting and solve following examples.

1. Piece Length: 1, 2, 3, 4, 5
Profit: 3, 5, 8, 9, 10
Rod Length = 8
2. Piece Length: 1, 3, 4, 6
Profit: 2, 7, 8, 20
Rod Length = 7
3. Piece Length: 1, 2, 3, 4
Profit: 5, 6, 8, 9
Rod Length = 5
4. Piece Length: 1, 2, 3, 4
Profit: 2, 5, 7, 8
Rod Length = 6
5. Piece Length: 1, 3, 4
Profit: 3, 8, 9
Rod Length = 5

2. Explain Matrix-Chain Multiplication and solve following examples.

1. Element : {5, 8, 4, 6, 2}
2. Element: {10, 20, 5, 15}
3. Element: {8, 3, 5, 7, 6}
4. Element: {6, 2, 9, 4, 7}
5. Element: { 4, 10, 3, 12, 20, 7}

3.Explain Elements of Dynamic Programming .

4.Explain Longest Common Subsequence (LCS) and solve following examples.

1. X = AGGTAB, Y = GXTXAYB
2. X = ABCDGH, Y = AEDFHR
3. X = ABCDEF, Y = FBDAMN
4. X = XMJYAUZ, Y = MZJAWXU
5. X = BACDB, Y = BDCAB

5. Explain Optimal Binary Search Trees (OBST) and solve following examples.

1. Keys = [5, 15, 25], Frequencies = [2, 3, 4].
2. Keys = [10, 20, 30], Frequencies = [4, 2, 6].
3. Keys = [10, 20, 30, 40], Frequencies = [3, 1, 4, 2].

4. Keys = [1, 2, 3, 4], Frequencies = [2, 5, 3, 4].

5. Keys = [8, 12, 16, 20], Frequencies = [1, 4, 2, 3].